

# SAMPLING THEORY AND TESTS OF SIGNIFICANCE

## Table of Content

|       |                                               |
|-------|-----------------------------------------------|
| 5.1   | Methods of Sampling                           |
| 5.1.1 | Sample Method                                 |
| 5.1.2 | Types of Data Sampling Methods                |
| 5.2   | Simple Random Sampling                        |
| 5.2.1 | Types of Simple Random Sampling               |
| 5.2.2 | Simple Random Sampling Formula                |
| 5.2.3 | Use Simple Random Sampling                    |
| 5.2.4 | Applications of Simple Random Sampling        |
| 5.2.5 | Purpose of Simple Random Sampling             |
| 5.2.6 | Characteristics of Simple Random Sampling     |
| 5.2.7 | Advantages of Simple Random Sampling          |
| 5.2.8 | Disadvantages of Simple Random Sampling       |
| 5.3   | Stratified Random sampling                    |
| 5.3.1 | Types of Stratified Random Sampling           |
| 5.3.2 | Benefits of Stratified Random Sampling        |
| 5.3.3 | Steps for Stratified Random Sampling          |
| 5.3.4 | Applications of Stratified Random Sampling    |
| 5.3.5 | When to use Stratified Random Sampling        |
| 5.3.6 | Characteristics of Stratified Random Sampling |
| 5.3.7 | Advantages of Stratified Random Sampling      |
| 5.3.8 | Disadvantages of Stratified Random Sampling   |
| 5.4   | Systematic Sampling                           |
| 5.4.1 | Types of Systematic Sampling                  |
| 5.4.2 | Advantages of Systematic Sampling             |
| 5.4.3 | Disadvantages of Systematic Sampling          |
| 5.4.4 | Steps to Create a Systematic Sample           |
| 5.5   | Tests of Significance                         |
| 5.5.1 | Significance Testing                          |
| 5.5.2 | Process of Significance Testing               |
| 5.5.3 | Types of Errors                               |
| 5.6   | Statistical Tests                             |
| 5.6.1 | Types of Statistical Tests                    |
| 5.7   | Z- test                                       |
| 5.8   | T- Test                                       |
| 5.9   | Chi-square test                               |
| 5.10  | F-Test                                        |

## 5.1 Methods of Sampling

The sampling method involves selecting a subset of individuals or observations from a larger population to collect data and make inferences about the entire population.

- It is a practical and efficient way to gather data when it is impractical or impossible to collect information from every member of the population.
- Different methods of sampling include Random Sampling, Purposive Sampling, Stratified Sampling, Systematic Sampling, Quota Sampling, and Convenience Sampling.

### 5.1.1 Sample Method:

- The **method of collecting data from a population, regarding a sample on a group of items and examining it to draw out some conclusion**, is known as **Sample Method**.
- This method is even used in the day-to-day lives of people.
- The sampling method of collecting data is suitable for a large population and when the investigator does not require a high level of accuracy.
- It is also preferred by investigators when they do not need an intensive examination of items.

[Data sampling](#) is a statistical method that involves selecting a part of a population of data to create representative samples. The fundamental aim is to draw conclusions about the entire population without having to engage with every individual data point, thus saving time, resources, and effort while still achieving accurate results.

### 5.1.2 Types of Data Sampling Methods

Sampling techniques are categorized into two main types: probability sampling and non-probability sampling. Each type is tailored to specific research needs and offers unique advantages and challenges.

#### 1. Probability Sampling

- Simple Random Sampling
- Stratified Sampling
- Cluster Sampling
- Systematic Sampling

#### 2. Non-Probability Sampling

- Convenience Sampling
- Purposive Sampling
- Snowball Sampling
- Quota Sampling

## 1. Probability Sampling Techniques

Probability sampling is defined by the principle that every member of the population has a known and equal chance of being selected. This method is critical for producing unbiased, representative samples.

### 1.a) Simple Random Sampling

**Simple random sampling** is the most straightforward probabilistic sampling technique.

- Every member of the population has an equal chance of being included in the sample, similar to a lottery.
- This method requires a complete list of the population, from which members are chosen randomly, either manually or using random number generation tools.
- Suppose you have a list of 10,000 voters in a town and you want to survey 1,000 of them.
- Each voter's name is assigned a number, and a random number generator picks 1,000 unique numbers corresponding to the voters who will be surveyed.

#### Simple Random Sampling: When to Use It

Simple random sampling is an appropriate choice under certain conditions or circumstances.

- **Homogeneous Population:** Ideal for populations with similar characteristics, ensuring equal selection.
- **Small Population Size:** Efficient for small, easily accessible populations.
- **Limited Resources:** Cost-effective and straightforward, ideal for studies with limited resources.
- **Equal Representation:** Ensures proportional representation of each subgroup in the sample.
- **Population Heterogeneity:** Provides unbiased representation of population diversity when the population exhibits homogeneity or random distribution.

### 1.b) Stratified Sampling

Stratified sampling is a method of dividing a population into subgroups based on characteristics like age, gender, or income level, and randomly selecting samples from each stratum proportionally. This ensures each subgroup is adequately represented in the sample, making it useful for specific subgroups of interest.

#### Stratified Sampling: When to Use It

- **Heterogeneous Population:** When the population exhibits variability in characteristics of interest.
- **Specific Subgroup Analysis:** When certain subgroups need to be represented accurately in the sample.

- **Increased Precision:** When precision is desired for estimates within each subgroup.

### 1.c) Cluster Sampling

Cluster sampling is a method of dividing a population into clusters, such as households or schools, and randomly selecting them for a sample. This method is efficient when obtaining a complete list of individuals, such as in a study on obesity prevalence in rural communities.

#### Cluster Sampling: When to Use It

- **Large Geographical Area:** When the population is spread out over a large geographical area.
- **Cost and Time Efficiency:** When it's more cost-effective and time-efficient to sample clusters rather than individuals.
- **Infeasibility of Complete Population List:** When obtaining a complete list of individuals in the population is impractical.

### 1.d) Systematic Sampling

Systematic sampling is a method of selecting individuals from a larger population at regular intervals, determined by dividing the population size by the desired sample size. For instance, in an employee directory of 500 people, every 10th person would be selected.

#### Systematic Sampling: When to Use It

Systematic sampling is particularly useful in several scenarios:

- **Efficient for Large Populations:** Systematic sampling allows researchers to cover a sizable portion of the population while maintaining randomness.
- **Ideal for Ordered Populations:** Systematic sampling ensures a representative sample of a population ordered by time, space, or another characteristic.
- **Balances Randomness and Efficiency:** Systematic sampling offers a balance between randomness and efficiency, making it a practical choice for real-world settings.
- **Accessibility and Convenience:** Systematic sampling allows for a straightforward selection process in easily accessible populations.
- **When Randomization is Difficult:** Systematic sampling provides a structured approach to sampling that is easier to implement and can still yield representative samples.

**2. Non-Probability Sampling Techniques** Non-probability sampling methods do not provide all the members of the population an equal chance of participating in the study. These techniques are used when the availability of a complete list is not possible, or when the research does not require a random sample.

### 2.a) Convenience Sampling

Convenience sampling is a method where researchers choose individuals who are easily accessible, rather than randomly chosen from the entire population, when other methods are impractical due to time, cost, or logistical constraints, but it may introduce bias and limit the generalizability of findings.

### **Convenience Sampling: When to use it?**

Convenience sampling is typically used in situations where:

- **Time Constraints:** Ideal for quick insights in limited time.
- **Cost Considerations:** Cost-effective due to minimal planning or resource requirements
- **Logistical Challenges:** Provides a practical solution for remote locations or restricted access. Offers data from readily accessible individuals, even if they don't represent the entire population.

### **2.b) Purposive Sampling**

Purposive sampling is a method used by researchers to select specific individuals or elements from a population based on predetermined criteria, aiming to gain insights into specific subgroups or phenomena. This method is guided by the researcher's expertise, facilitating in-depth exploration.

#### **Purposive Sampling: When to use it?**

Purposive sampling is employed under the following circumstances:

- **Specific Research Objectives:** Ideal for targeting specific characteristics or subgroups within the population.
- **Expertise or Prior Knowledge:** Used when researchers have prior knowledge about the population.
- **In-depth Exploration:** Beneficial for exploring specific phenomena or behaviors in depth. Targets individuals exhibiting specific traits or behaviors of interest for detailed insights.

### **2.c) Snowball Sampling**

Snowball sampling is a non-probability method where participants are recruited through referrals, used to study hard-to-reach or hidden populations, such as marginalized communities or individuals with rare characteristics, and is beneficial in situations where traditional methods are impractical.

#### **Snowball Sampling: When to use it?**

Snowball sampling is appropriate in the following scenarios:

- **Hard-to-Reach Populations:** Snowball sampling is a valuable method for studying hard-to-reach populations, leveraging existing connections within the target population to recruit additional participants.
- **Lack of Sampling Frame:** Snowball sampling is a practical method that uses referrals from existing participants to identify and recruit new participants in populations without clear boundaries or sampling frames.

- **Sensitive Topics:** Snowball sampling is ideal for sensitive research, as it involves participants recruited through personal networks who are more likely to trust the researcher and share their experiences.

## 2.d) Quota Sampling

Quota sampling is a method used in market research, opinion polling, and social sciences to select a sample based on predetermined quotas to ensure proportional representation of key population segments.

### Quota Sampling: When to use it?

Quota sampling is applicable under the following conditions:

- **Representation of Key Characteristics:** Sets predetermined quotas for various population segments to ensure proportional representation of key characteristics.
- **Practical Constraints on Random Sampling:** Provides a practical alternative to random sampling when logistical constraints or unavailability of a sampling frame prevent random sampling.
- **Market Research and Opinion Polling:** Employed in these fields to obtain samples that mirror the demographic composition of the target population.

### Best Practices for Choosing Data Sampling Methods

Deciding on the type of sampling to use depends on several factors, including the research objectives, characteristics of the population, available resources, and practical constraints.

- **Define research objectives:** Clarify specific research goals and questions.
- **Understand population characteristics:** Understand population size, diversity, distribution, and homogeneity.
- **Assess sampling frame availability:** Determine if a comprehensive population representation is available.
- **Consider budget and time constraints:** Evaluate available resources for the sampling process.
- **Evaluate sampling precision:** Determine the level of precision needed for the study.
- **Explore sampling methods:** Familiarize with probability and non-probability sampling.
- **Match method to objectives and constraints:** Select the method that aligns with research objectives, population characteristics, resources, and constraints.
- **Consider sampling bias and error:** Assess potential sources and minimize them through design and validation processes.
- **Conduct pilot test if necessary:** Evaluate effectiveness and feasibility before implementing on a larger scale.

- **Document decision-making process:** Ensure transparency and reproducibility of the research process.

## Simple Random Sampling

9 3 4 6 5 1 5 2 0 6 4 4 9 2 1 6 3 4 5 5  
 0 8 0 0 0 9 4 8 6 2 7 9 3 6 0 0 3 4 4 6  
 9 1 5 7 1 1 2 9 0 1 0 4 2 9 5 0 1 7 1 2  
 7 7 1 0 3 2 2 9 3 1 5 1 0 2 8 0 1 5 3 8  
 4 6 5 9 7 9 2 8 1 5 0 7 2 0 5 3 2 7 7 4  
 5 4 3 6 0 3 5 4 4 0 5 1 6 3 5 4 7 5 8 2  
 8 2 9 2 0 7 1 3 8 0 3 7 8 8 8 6 3 2 9 5  
 4 7 9 2 3 2 8 7 6 3 2 9 7 6 9 5 5 7 3 8  
 0 6 9 6 4 6 3 9 5 4 6 9 8 4 3 7 3 8 1 7  
 6 3 6 3 9 3 8 4 1 9 0 3 8 2 7 6 0 7 3 9  
 0 9 2 1 2 2 5 6 9 4 6 8 2 8 7 0 4 6 3 6  
 1 7 5 9 5 9 8 9 9 3 9 7 5 2 9 9 6 9 5 7  
 3 9 7 6 9 1 5 6 7 9 4 9 7 7 5 3 2 5 0 6  
 6 7 8 7 8 5 1 0 1 1 8 6 9 3 2 6 1 2 5 1  
 4 2 7 4 2 1 2 6 0 1 8 1 1 7 1 8 0 5 1 1  
 1 3 6 9 9 7 7 3 8 4 3 4 6 5 3 3 2 6 0 0  
 5 1 2 5 2 6 2 8 2 1 5 8 1 3 9 5 2 0 6 2  
 5 8 2 6 9 8 1 8 0 5 4 7 1 0 0 5 0 5 3  
 5 6 6 5 9 6 1 7 2 0 5 2 0 0 5 4 1 6 8 7  
 7 5 3 8 1 8 0 9 2 1 1 4 6 7 4 5 0 1 1 4



## 5.2 Simple Random Sampling

**Definition:** Simple Random Sampling is a type of probability sampling for selecting a random sample from a population, in which each member of the population has an equal chance of being selected.

This means that every member of the population has the same probability of being chosen as any other member, and each possible sample of a given size has an equal probability of being selected.

### 5.2.1 Types of Simple Random Sampling

Types of Simple Random Sampling are as follows:

- **Simple random sampling with replacement:** Each member has an equal chance of selection. Once selected, the member isn't removed from the population.
- **Simple random sampling without replacement:** Each member can only be selected once.
- **Stratified random sampling:** Divides the population into subgroups based on characteristics, selects a simple random sample from each stratum.

**Sampling with replacement – independent events**

**Sampling without replacement – dependent events**

1. **Dependent:** occurrence of one event affect the next event.  

$$P(A \text{ and } B) = P(A) \times P(B|A)$$
2. **Independent:** occurrence of one event does not affect the other.

$$P(A \text{ and } B) = P(A) \times P(B)$$

**In general:**  $P(A \text{ and } B \text{ and } C \dots) = P(A) \times P(B) \times P(C) \dots$  if A, B, C are independent.

**Ex1.** A fair coin is tossed 5 times. Find  $P(\text{first 3 tosses are heads})$ .  
 $F$  (first 4 are heads),  $P(\text{at least one head in 4 tosses})$ .

Coin tosses are independent events.

- a)  $P(\text{First 3 tosses are heads}) =$  Since tosses are independent,  
 $P(3 \text{ heads}) = 0.5^3 = 0.125$ .
- b)  $P(\text{First 4 tosses are heads}) =$  Since tosses are independent,  
 $P(4 \text{ heads}) = 0.5^4 = 0.0625$   
 $P(\text{at least one head in 4 tosses}) = 1 - P(\text{all 4 heads are tails})$   
 $= 1 - 0.5^4 = 0.9375$

**Ex2.** In a group of 300 all adults, 272 are right-handed, 3 adults are selected with replacement. Find  $P(\text{all 3 are right-handed})$  and  $P(\text{all 3 left-handed})$ .

**Ans:**

$$P(\text{one right-handed}) = 272/300 = 0.907$$

$$P(\text{one left-handed}) = (300-272)/300 = 28/300 = 0.093$$

$$P(3 \text{ right-handed}) = (272/300)(272/300)(272/300) = (272/300)^3 = 0.745$$

$$P(3 \text{ left-handed}) = (28/300)(28/300)(28/300) = (28/300)^3 = 0.0008$$

$$P(\text{At least one right-handed}) = 1 - P(\text{All 3 left-handed})$$

$$= 1 - 0.0008 = 0.9992$$

**Ex3.** In a jar with 5 red, 6 blue and 2 white marbles. Two marbles are selected, find the probability that both are red if:

- a) If two marbles are selected with replacement.
- b) If two marbles are selected without replacement.

**Ans:**

- a) If marbles are replaced, the events are independent.  
 $P(\text{both are red}) = 5/13 * 5/13 = 0.1479$
- b) If two marbles are selected without replacement, the events are dependent,  
 $P(\text{both are red}) = 5/13 * 4/12 = 1282$ . Treating Sampling without replacement as independent if one of the following are satisfied:
  1. Sampling with replacement – independent events
  2. Sampling without replacement – dependent events
- 1. Assume a very big population when population size is not given. Only  $P(A)$  is given.



2. Use 5% guideline for cumbersome calculations: When sampling without replacement and the sample size is no more than 5% of the size of population, treat sampling as independent. (Even though they are actually dependent.)

**Ex1.** Assume that 10% of adults in the United States are left-handed. Find the probability that three selected adults all are left-handed.

Since the population size is not given only  $P(L) = 0.1$  is given, we can treat sampling without replacement as independent.

$$P(L \text{ and } L \text{ and } L) = P(L) \times P(L) \times P(L) = 0.1 \times 0.1 \times 0.1 \\ = 0.001$$

**Ex2.** In batch of 6400 light bulbs, 80 are defective.

If 12 light bulbs are selected from the batch without replacements, find probability that all are good.

**Ans:**

sample size = 12, population size = 6400

Since sampling proportion =  $12/6400 = 0.00188 < 0.05$

we can treat the sampling as independent by apply 5% guideline for cumbersome calculation.

$P(\text{one good}) = (6400-80)/6400 = 0.9875$

$P(\text{one defective}) = 80/6400 = 0.0125$

$P(\text{all 12 are good}) = 0.9875^{12} = 0.860$

$P(\text{at least one defective}) = 1 - P(\text{all 12 are good}) = 0.14$

## 5.2.2 Simple Random Sampling Formula

The formula for simple random sampling is as follows:

$$n/N = k$$

where,

$n$  is the sample size

$N$  is the population size

$k$  is the sampling fraction, which is the ratio of the sample size to the population size

To calculate the sample size, you can rearrange the formula as follows:

$$n = N \times k$$

For example, if you have a population of 1000 and you want to take a simple random sample with a sampling fraction of 0.10, the formula would be:

$$n = 1000 \times 0.10 = 100$$

Therefore, you would need to randomly select 100 individuals from the population to create your sample.

### Simple Random Sampling Method Steps:

- **Define the population:** Ensure the population is well-defined and includes all individuals to be studied.
- **Determine the sample size:** Choose a sample size that is large enough for accurate results but manageable.
- **Assign a number to each member of the population:** Use a random number generator or manually assign numbers.
- **Generate random numbers:** Use a random number generator to generate a set of random numbers corresponding to the sample size.
- **Select the sample:** Use the random numbers to select individuals from the population.
- **Analyze the sample:** Analyze the collected data to draw conclusions about the population.

### Examples of Simple Random Sampling

Here are some examples of simple random sampling:

- **Polling:** Researchers can use random sampling to determine a political candidate's approval rating. This ensures each voter has an equal chance of being selected.
- **Quality Control:** A random sample from the production line can be used to test product quality. This ensures the sample is representative of the entire production process.
- **Medical Research:** Random selection of individuals from the population can provide an unbiased estimate of the disease's prevalence.
- **Education Research:** Random selection of students from a school can help study the effectiveness of a new teaching method.
- **Social Science Research:** A representative sample of individuals can be selected and surveyed to collect data on attitudes and opinions towards a social issue.

### Simple Random Sampling Example Situation

Simple Random Sampling Example Situation is as follows:

A company wants to conduct a survey to understand the job satisfaction of its employees. The company has a total of 500 employees, and they want to select a sample of 50 employees using simple random sampling.

Formula: The formula for simple random sampling is:

$$n/N * (N-n)/(N-1)$$

where,

n: the sample size

N: the population size

In this case, n = 50 and N = 500.

So, the formula for calculating the probability of selecting a simple random sample of 50 employees from a population of 500 is:

$$50/500 * (500-50)/(500-1) = 0.1 * 0.902 = 0.0902, \text{ or } 9.02\%.$$

Therefore, the probability of selecting a sample of 50 employees using simple random sampling is 9.02%. This means that any set of 50 employees has an equal chance of being selected, and the sample is representative of the population.

### 5.2.3 Use Simple Random Sampling

Here are some situations where simple random sampling may be appropriate:

- **When the population is relatively small:** Simple random sampling is an ideal method to use when the population size is small because it is easy to implement and doesn't require a lot of resources.
- **When the population is homogenous:** Simple random sampling is suitable when the population is relatively homogenous, meaning that the characteristics of the individuals or elements are relatively similar.
- **When the researcher wants to avoid potential bias:** Simple random sampling is an excellent method to use when the researcher wants to avoid any potential bias in the sample selection process.
- **When the data is not strongly correlated:** Simple random sampling is appropriate when the data is not strongly correlated. Strong correlation between variables can result in an unrepresentative sample.

### 5.2.4 Applications of Simple Random Sampling

Simple random sampling has numerous applications in various fields, including:

- **Public opinion polls:** Simple random sampling is widely used in public opinion polls to collect data from a representative sample of the population.
- **Medical research:** In medical research, simple random sampling is used to select participants for clinical trials or to conduct surveys to collect data on health-related issues.
- **Quality control:** Simple random sampling is used in quality control to select products or items from a production line for inspection to ensure that they meet certain quality standards.
- **Education research:** Simple random sampling is used in educational research to select students or schools to participate in studies.
- **Market research:** Simple random sampling is used in market research to collect data on consumer behavior and preferences.

- **Environmental studies:** Simple random sampling is used in environmental studies to collect data on various environmental factors.
- **Social science research:** Simple random sampling is widely used in social science research to study various social phenomena.

### 5.2.5 Purpose of Simple Random Sampling

- Aims to obtain a representative sample of a population.
- A representative sample accurately reflects the entire population's characteristics.
- Ensures equal chance of selection for every individual or element.
- Eliminates potential biases from non-random sampling methods.
- Helps researchers make generalizations about the population with high confidence.
- Results of statistical analyses can be used to make inferences about the larger population.

### 5.2.6 Characteristics of Simple Random Sampling

- **Randomness:** Ensures equal chance of selection for each element in the population, reducing potential bias.
- **Independence:** Selection of each element is independent, preventing influence on others.
- **Objectivity:** Selection based solely on chance, no subjective judgment or influence.
- **Equal probability:** Every possible sample of a given size has an equal chance of selection.
- **Simple to use:** Easy to understand, implement, and analyze, making it a common sampling method in statistical research.
- **Applicable for any population:** Applicable to any population, as long as complete list of all elements is available.

### 5.2.7 Advantages of Simple Random Sampling

Simple random sampling has several advantages over other sampling methods, including:

- **Representativeness:** Ensures equal chance of selection for every individual in the population.
- **Unbiased:** Eliminates potential biases from non-random sampling methods.
- **Easy to use:** Easy to understand, implement, and analyze, suitable for researchers with limited resources.

- **Application of probability theory:** Allows for estimation of margin of error and confidence interval.
- **Flexibility:** Suitable for any population, as long as a complete list of all elements is available.

### 5.2.8 Disadvantages of Simple Random Sampling

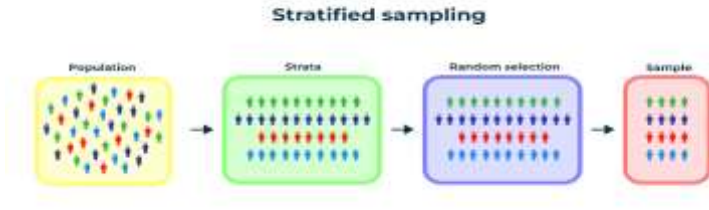
While simple random sampling has several advantages, it also has some disadvantages, including:

- **Large sample size:** Requires a large sample size, increasing research costs and time.
- **Inefficiency:** Inefficient for large, geographically dispersed populations.
- **Under-representation:** Can lead to under-representation of certain subgroups
- **Selection bias:** Doesn't eliminate other biases like measurement or response bias.
- **Not suitable for small populations:** Small sample size may not be representative.

### 5.3 Stratified Random sampling

Stratified Random Sampling is a technique in Machine Learning and Data Science that selects random samples from a large population to avoid bias and sampling errors. It divides the population into smaller sub-groups and randomly selects samples, making it more accurate than traditional random sampling.

- Stratified Random Sampling splits the entire population into smaller subsets, known as Strata, based on a specific characteristic in the data.
- This process of random sampling from Strata is carried out, ensuring all characteristics or features in the entire population are reflected in the sample dataset.
- This approach eliminates the possibility of sampling error due to population bias, making the Machine Learning model more accurate.



Stratified random sampling, in simple terms, the entire Stratified Random Sampling consists of two main steps –

- **Forming Strata** – Filtering out the values from a dataset based on their features and forming smaller sub-groups.
- **Random Sampling from the Strata** – Randomly pick samples from those sub-groups to form the sample dataset, without considering the entire population.

### 5.3.1 Types of Stratified Random Sampling:

Mainly, there are two types Stratified Random Sampling possible –

1. **Proportionated Stratified Random Sampling**
2. **Disproportionated Stratified Random Sampling**

#### **Proportionated Stratified Random Sampling –**

Proportionate stratified random sampling is a type of stratified random sampling where the number of random samples taken from each stratum is proportional to its size in the population.

- This method is commonly used when the strata are relatively homogeneous, ensuring a representative sample.
- However, it may not be as efficient if some strata are smaller than others.

**Example:** Surveying student satisfaction in a university with freshmen, sophomores, juniors, and seniors.

**Disproportionate stratified random sampling:** is a method used to oversample or under sample strata based on their size in the population,

- often used in heterogeneous strata or when certain strata are considered more important.
- While more efficient than proportionate stratified random sampling, it may not be representative of the entire population.

**Example:** Surveying residents' opinions on a public transportation system in three districts with different population sizes.

### 5.3.2 Benefits of Stratified Random Sampling:

The benefits of Stratified Random Sampling are,

1. **Improved Precision:** Divides population into smaller subgroups called "strata" based on feature characteristics. Ensures equal representation of all features in the sample dataset.
2. **Enhanced Comparisons:** Enables comparison of each element's characteristics, ensuring accurate and bias-less Machine Learning models.
3. **Resource Efficiency:** Allocating resources efficiently by dividing population into strata, allowing for more emphasis on specific subgroups.

### 5.3.3 Steps for Stratified Random Sampling:

Now, we will see, how we could perform Stratified Random Sampling, in a stepwise manner.

- **Step 1: Define Population and Subgroups.**
  - Identify and select unique characteristics for population division and strata formation.
  - Choose clear, distinct features to differentiate between subgroups.
  - Use multiple columns/features to stratify the dataset and create sub-groups.
- **Step 2: Separate the Population into Strata.**
  - Consider each member of the population and add them to different strata based on their characteristic and unique feature.
  - The collection of all strata is known as strata.
- **Step 3: Decide on the Sample Size for Each stratum**
  - Decide on the type of Stratified Random Sampling (Proportionate or Disproportionate) to be used.
  - The sample size should be large enough to ensure equal representation of data from each stratum in the sample dataset.
- **Step 4: Randomly Sample from Each Stratum**
  - Use the random sampling method to collect data from each stratum and form the sample dataset.
  - Combine all samples into one representative sample by concatenating them.

### 5.3.4 Applications of Stratified Random Sampling

- **Market Research:** Divides consumer feedback into categories for efficient analysis.
- **Educational Research:** Divides student data into strata based on class, section, age for academic performance analysis.

- **Health Studies:** Divides population into strata based on age, gender, common disease symptoms for quicker diagnosis and treatment.

### 5.3.5 When to use Stratified Random Sampling:

There are certain scenarios in which Stratified Random Sampling will work better than that of the simple Random Sampling method. Some of them are listed below –

- **Heterogeneous Population:** Stratified random sampling is more effective in diverse, complex populations with numerous unique features.
- **Significant Variability:** Random sampling may not accurately represent the variability of a population, leading to potential sampling bias.
- **Comparative Analysis:** Stratified random sampling can be used to divide survey data into separate subgroups for equal representation.
- **Improved Precision:** Stratified random sampling eliminates sampling bias, enhancing the precision of machine learning models.
- **Population Research:** Ideal for epidemiology, where representative data of various demographic groups is needed.

### 5.3.6 Characteristics of Stratified Random Sampling

1. **Division into Strata** – The population is divided into sub-groups known as “strata” based on the features/ characteristics of them.
2. **Homogeneity Within Strata** – Strata are formed in such a manner that all the elements of each stratum are homogeneous or share the same unique characteristics, using which they can be independently identified.
3. **Random Sampling Within Strata** – In every stratum, Random Sampling is done to ensure that each stratum contributes equally in forming the sample dataset.
4. **Statistical Weighting** – After forming the sample dataset, different statistical approaches can be taken to measure the contribution of each of the stratum in forming the sample dataset.
5. **Increased Precision** – The aim of Stratified Sampling is to increase the precision of the Machine Learning model which will be created using that sample dataset.

### 5.3.7 Advantages of Stratified Random Sampling

- **Improves Population Representation:** Divides the entire population based on unique features, ensuring all features are reflected in the Sample Dataset.



- **Reduces Sampling Error:** Ensures all unique features are reflected proportionately in the Sample Dataset.
- **Increases Efficiency:** Allows for resource allocation based on the importance of each strata.
- **Reduces Bias:** Prevents under sampling or oversampling, especially for minority groups or unique characteristics.
- **Increases Precision:** Each stratum is sampled independently, maximizing the precision of the Machine Learning model and the Sample Dataset.
- **Promotes Generalizability:** Sample datasets made using Stratified Random Sampling generalize the entire population and its unique features proportionately.

### 5.3.8 Disadvantages of Stratified Random Sampling

- **Complexity:** Stratified Random Sampling is more complex than Simple Random Sampling due to the need for specific characteristics of the population. This complexity can lead to issues if not properly stratified or the population is not diverse.
- **Selection of Stratification Feature:** Proper stratification of the data is crucial to ensure the sample dataset accurately represents the population.
- **Resource Intensive:** Large populations with unique features can create large strata and large sample sizes, requiring significant resources and time.
- **Over stratification:** Over-diversification can lead to features not being entirely unique or having similarities, making it difficult to gather adequate data.
- **Practical Challenges:** The approach may be difficult to execute due to the lesser variety in the population or poorly defined unique features.
- **Relation between the strata:** Hidden or underlying relationships between features or strata can lead to misleading results.

## 5.4 Systematic Sampling

Systematic Sampling is a probability sampling method that randomly selects sample members from a large population with a fixed interval, allowing researchers, particularly marketing and sales professionals, to obtain representative findings without contact with every individual.

### Systematic Sampling Overview

- Involves structured selection of items from a list.
- Best for groups arranged in order.
- Efficient and systematic strategy compared to random sampling.

- Crucial interval between selected items ensures fair representation.
- Starts at a random point to prevent un-intentional biases.
- Ensures evenly distributed samples throughout the population.

### 5.4.1 Types of Systematic Sampling

Systematic Sampling, a method for selecting representative samples from larger populations, comes in three main types, each with its unique approach:

Some of the types of Systematic Sampling are:

- **Linear Systematic Sampling:** In this method, individuals are chosen at regular intervals from a list. For instance, every 5th person is selected from a given population list after a random starting point.
- **Circular Systematic Sampling:** Particularly useful for circular or sequential data, this method involves selecting samples at regular intervals, looping back to the beginning when reaching the end of the sequence.
- **Stratified Systematic Sampling:** The population is divided into distinct strata or subgroups based on specific criteria. Then, systematic sampling is applied within each stratum independently to ensure representation from all groups.
- **Cluster Systematic Sampling:** Here, the population is divided into clusters, and a systematic sample of clusters is chosen. Then, within the selected clusters, all individuals are sampled systematically.

### 5.4.2 Advantages of Systematic Sampling

1. **Easy to Understand and Use:** Systematic sampling follows a set pattern, making it simple for researchers or surveyors.
2. **Saves Time:** Systematic sampling is quicker than individual selection, making it ideal for time-sensitive research or resource constraints.
3. **Keeps the Sample Representative:** It ensures equal chance of selection for all group members, ensuring a representative sample.
4. **Reduces Differences in the Sample:** Regular pattern reduces risk of over-focusing on one part of the group, resulting in a balanced representation.
5. **Cost-Friendly:** Systematic sampling is less time-consuming and requires less training, making it a budget-friendly option.

### 5.4.3 Disadvantages of Systematic Sampling

1. **Sensitive to Patterns:** Systematic sampling can unintentionally pick up on patterns in the population, especially if the population has repeating trends.
2. **Risk of Skewed Results:** If the sample selection doesn't match the population's order, results may be skewed.
3. **Not Good for Irregular Groups:** Systematic sampling may miss areas where the population is scattered unevenly, providing an incomplete picture.
4. **Introduces Bias:** Systematic sampling may introduce bias if there's a hidden order or trend in the population.
5. **Not Great with Unknown Patterns:** Systematic sampling assumes a regular sequence, which may not accurately represent the whole group.

### 5.4.4 Steps to Create a Systematic Sample

Systematic Sampling is a method used in statistics to choose a representative sample from a larger group, or population.

1. **Defining Population:** The entire group of things or people to be studied is the population.
2. **Determining Sample Size:** The sample size is determined by the number of things or people to be studied. **For example**, 100 students out of 1000 are chosen.
3. **Calculating Interval (k):** The sampling interval (k) is determined to ensure fair selection of students. **For example**, if 1000 students are chosen, the interval would be 1000 divided by 100, resulting in every 10th student in the sample.
4. **Random Start:** To ensure unbiased sample selection, a starting point is randomly chosen.
5. **Selecting Every kth Element:** After a starting point, each student is selected, starting from the 10th until the desired sample size is reached.

### Examples of Systematic Sampling

Imagine there's a corporation with 15,000 employees, and the Human Resources department wants to gauge employee satisfaction. They decide to use systematic sampling with the following parameters:

**Population:** All 15,000 employees in the corporation.

**Desired Sample Size:** They aim to survey 300 employees. Now, let's determine the systematic sampling process:

- **Calculate Interval (k):** The sampling interval is calculated as 15,000 (total employees) divided by 300 (sample size), which equals 50.

- **Random Start:** They randomly select a starting point by choosing any employee between the 1st and 50th in the employee list.
- **Select every  $k^{\text{th}}$  Employee:** Starting from the chosen employee, they survey every 50th employee (50th, 100th, 150th, and so on) until they reach the desired sample size of 300.

This approach ensures that every 50th employee has an equal chance of being included, providing a representative sample of employee satisfaction within the corporation.

## 5.5 Tests of Significance

Tests of significance are statistical tests used to compare observed data with a hypothesis, assessing its truth in further analysis. They are crucial in descriptive statistical research to reveal the truth, ensuring good quality samples, minimal errors, and precise measures. The sample size, which determines the probability of an event occurring without proper research, can be weak or strong depending on statistical significance. Careless language in experiment reports can lead to misinterpreting the study's significance.

### 5.5.1 Significance Testing

- Involves assessing the significance of experimental results.
- Quantitative significance tests are defined and relevant.
- Simple significance tests are used to identify certain levels of error.
- Trial designers may predefine sampling error probability.
- Population sampling tests do not study the whole, resulting in inherent sampling error.
- Testing significance is a crucial part of statistical research.

#### Null Hypothesis

Null Hypothesis in Tests

- Starts with a null hypothesis ( $H_0$ ).
- $H_0$  represents a theory suggested but not proven.

**Example:** In a clinical test of a replacement drug,  $H_0$ : No difference between the two drugs on average..

### 5.5.2 Process of Significance Testing

In the process of testing for statistical significance, the following steps must be taken:

**Step 1:** Start by coming up with a research idea or question for your thesis.

**Step 2:** Create a neutral comparison to test against your hypothesis.

**Step 3:** Decide on the level of certainty you need for your results,

which affects the type of sign language translators and communication methods you'll use.

**Step 4:** Choose the appropriate statistical test to analyze your data accurately.

**Step 5:** Understand and explain what your results mean in the context of your research question.

### 5.5.3 Types of Errors

There are basically two types of errors:

- Type I Error
- Type II Error

#### Type I Error:

- Researchers find relationship presumed to be a case, but evidence shows it's not a function explained.
- Leads to failure of the researcher, accepting the null hypothesis instead of rejecting it.
- Identified with  $\alpha$  (alpha) as the probability of the error.

#### Type II Error:

- Similar to Type I error, researchers suppress emotions and avoid experiencing connection.
- Expected to see the research hypothesis as true and treat null hypotheses as false.
- Identified with  $\beta$  as the possibility of making a type II error, an error of omission.

## 5.6 Statistical Tests

- One-tailed tests determine the significance of a finding in a data set.
- One-tailed tests consider changes in a parameter from a baseline.
- Two-tailed tests consider changes that could go in either direction from the baseline.
- For instance, a two-tailed test would be used to determine if a new teaching method improves or worsens test scores.

### 5.6.1 Types of Statistical Tests

- Uses one-tailed or two-tailed statistical tests to determine the statistical significance of a parameter.
- Also known as lateral flow and dipstick tests.
- One-tailed test makes parameter estimation differences within one side plausible.
- Two-tailed test considers deviations from both sides of benchmark value.
- "Tail" refers to extreme points of the distribution where outliers reject the null hypothesis.

- The application of one-tailed or two-tailed test depends on the research hypothesis.

### **What is p-Value Testing:**

- P-value is a significant term in hypothesis testing.
- It's a function determining the threshold value before starting the test.
- The significance level is typically 1% or 5%.
- The Significance function parameter is  $\alpha$ .
- If p-value is greater than or equal to  $\alpha$ , inconsistency exists between the null model and data.
- The null hypothesis should be rejected and a new hypothesis may be assumed.

### **T-Test for Medical Research – The T Test**

- T-Test is used in medical research to determine the statistical significance of a difference between two assumed normal populations.
- The  $H_0$  (null hypothesis) suggests no significant difference in blood pressure between the two groups.
- The  $H_{A1}$  (alternative hypothesis) indicates a significant difference.
- The t-test helps to check if outcomes are significantly different, reducing the possibility of confusing hypotheses.

### **Chi-Square Analysis in Market Research**

#### **Market Research on Customer Satisfaction and Product Preferences**

- Aimed to determine the link between customer satisfaction and product preferences.
- Utilized chi-square test to assess the association between the two variables.
- $H_0$  null hypothesis suggests no relationship between customer satisfaction and product preferences.
- $H_1$  alternative hypothesis suggests a correlation between customer satisfaction and product preferences.
- The chi-square test helped determine the statistical significance of customer satisfaction and product preferences.
- Concluded that customer satisfaction affects the market conception of goods for the target market.

### **ANOVA in Educational Research**

Studying differences between different learning methods and their impact on student achievement.

- **Null Hypothesis (HO):** No differences in scores between groups.
- **Alternative Hypothesis (HA):** At least one group has a different mean.
- **Analysis of Variance (ANOVA):** Determines statistically significant difference in performance across teaching methods.

### **Regression Analysis in Economics**

- Examines the link between ad cost and revenue in businesses.
- Null space suggests no linear connection between ad spending and purchases.
- Regression analysis used to determine if sales changes are attributed to advertising changes.

### **Paired T-Test in Psychology**

- Psychologists study new therapy for anxiety relief.
- Patients' anxiety levels evaluated before and after intervention.
- Null hypothesis: No significant difference in anxiety levels post-intervention.
- Paired t-test used to statistically prove observed change in anxiety scores.

## **5.7 Z-test**

The Z-test is useful for large sample sizes and standard deviations. Statistical tests compare samples or groups, examining the probability of acquiring observed data under specific hypotheses, offering a framework for evaluating evidence.

- Statistical test determining significant difference between sample mean and known population mean.
- Useful when population standard deviation is known.
- Particularly useful for large sample sizes (>30).
- Determines if test statistics can be approximated using normal distribution.
- Determines if two sample means are approximately the same or different.

The Z-test calculates the standard deviation of the sample mean from the population mean, resulting in a Z-score, also known as Z-Statistics.

$$\text{Z-Score} = \frac{\bar{x} - \mu}{\sigma}$$

Where,

$\bar{x}$  : mean of the sample.

$\mu$  : mean of the population.

$\sigma$  : Standard deviation of the population.

**z-test assumes that the test statistic (z-score) follows a standard normal distribution.**

### Example

The average family annual income in India is 200k, with a standard deviation of 5k, and the average family annual income in Delhi is 300k. Then Z-Score for Delhi will be.

$$\begin{aligned}\text{Z-Score} &= \frac{\bar{x} - \mu}{\sigma} \\ &= \frac{300 - 200}{5} \\ &= 20\end{aligned}$$

This indicates that the average family's annual income in Delhi is 20 standard deviations above the mean of the population (India).

The Z-test is used when the sample size is greater than 30, and samples are drawn randomly from the population. The standard deviation of the population is known, and the data should be normally distributed, with a normal distribution assumed for large sample sizes.

Hypothesis testing is a method to validate experimental claims.

- **Null Hypothesis:** A statement that a population parameter's value equals the claimed value.
- **Alternative Hypothesis:** A statement that the parameter's value differs from the claimed value.
- **Level of Significance:** The degree of significance in which the null hypothesis is accepted or rejected.
- **Alpha ( $\alpha$ )** is the level of significance chosen due to the inability to achieve 100% accuracy in most experiments.

### 5.7.1 Steps to perform Z-test

- Identify null and alternate hypotheses.
- Determine significance level ( $\alpha$ ).
- Find critical value of z.
- Calculate z-test statistics using formula:  $z =$

$$Z = \frac{(\bar{x} - \mu)}{(\sigma / \sqrt{n})}$$

where,

$\bar{x}$  : mean of the sample.

$\mu$  : mean of the population.

$\sigma$  : Standard deviation of the population.

n: sample size.

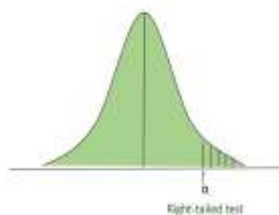
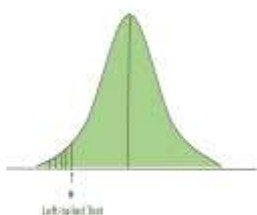
- Calculate sample mean, population mean, standard deviation, and sample size.



- Compare with hypothesis and decide to reject null hypothesis.

### 5.7.2 Type of Z-test

1. **Left-tailed Test:** In this test, our region of rejection is located to the extreme left of the distribution. Here our null hypothesis is that the claimed value is less than or equal to the mean population value.
2. **Right-tailed Test:** In this test, our region of rejection is located to the extreme right of the distribution. Here our null hypothesis is that the claimed value is less than or equal to the mean population value.



### 3. One-Tailed Test

A school claimed that the students who study that are more intelligent than the average school. On calculating the IQ scores of 50 students, the average turns out to be 110. The mean of the population IQ is 100 and the standard deviation is 15. State whether the claim of the principal is right or not at a 5% significance level.

First, we define the null hypothesis and the alternate hypothesis. Our null hypothesis will be:

$$H_0: \mu = 100$$

and our alternate hypothesis.

$$H_0: \mu > 100$$

State the level of significance. Here, our level of significance is given in this question ( $\alpha = 0.05$ ), if not given then we take  $\alpha = 0.05$  in general.

Now, we compute the Z-Score:

$$X = 110$$

$$\text{Mean} = 100$$

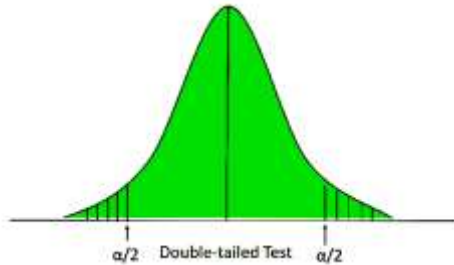
$$\text{Standard Deviation} = 15$$

$$\text{Number of samples} = 50$$

$$\begin{aligned}
 \text{Z-Score} &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\
 &= \frac{110 - 100}{15 / \sqrt{50}} \\
 &= \frac{10}{2.12} \\
 &= 4.71
 \end{aligned}$$

Now, we look up to the z-table. For the value of  $\alpha=0.05$ , the z-score for the right-tailed test is 1.645. Here  $4.71 > 1.645$ , so we reject the null hypothesis. If the z-test statistics are less than the z-score, then we will not reject the null hypothesis.

4. **Two-tailed test:** In this test, our region of rejection is located to both extremes of the distribution. Here our null hypothesis is that the claimed value is equal to the mean population value.



#### Two-sampled z-test

In this test, we have provided 2 normally distributed and independent populations, and we have drawn samples at random from both populations. Here, we consider  $\mu_1$  and  $\mu_2$  to be the population mean, and  $\bar{X}_1$  and  $\bar{X}_2$  to be the observed sample mean. Here, our null hypothesis could be like this:

$H_0: \mu_1 - \mu_2 = 0$  and alternative hypothesis  $H_1: \mu_1 - \mu_2 \neq 0$   
and the formula for calculating the z-test score:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Where,  $\sigma_1$  and  $\sigma_2$  are the standard deviation and  $n_1$  and  $n_2$  are the sample size of population corresponding to  $u_1$  and  $u_2$ .

Example:

There are two groups of students preparing for a competition: Group A and Group B. Group A has studied offline classes, while Group B has studied online classes. After the examination, the score of each student comes. Now we want to determine whether the online or offline classes are better.

Group A: Sample size = 50, Sample mean = 75, Sample standard deviation = 10

Group B: Sample size = 60, Sample mean = 80, Sample standard deviation = 12

Assuming a 5% significance level, perform a two-sample z-test to determine if there is a significant difference between the online and offline classes.

**Solution:**

Step 1: Null & Alternate Hypothesis

Null Hypothesis: There is no significant difference between the mean score between the online and offline classes

$$\mu_1 - \mu_2 = 0$$

Alternate Hypothesis: There is a significant difference in the mean scores between the online and offline classes.

$$\mu_1 - \mu_2 \neq 0$$

Step 2: Significance Label

Significance Label: 5%

$$\alpha = 0.05$$

Step 3: Z-Score

$$\begin{aligned} \text{Z-score} &= \frac{(x_1 - x_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \\ &= \frac{(75 - 80) - 0}{\sqrt{\frac{10^2}{50} + \frac{12^2}{60}}} \\ &= \frac{-5}{\sqrt{2 + 2.4}} \\ &= \frac{-5}{2.0976} \\ &= -2.384 \end{aligned}$$

Step 4: Check to Critical Z-Score value in the Z-Table for  $\alpha/2 = 0.025$

Critical Z-Score = 1.96

**Step 5:** Compare with the absolute Z-Score value

absolute(Z-Score) > Critical Z-Score

Reject the null hypothesis. There is a significant difference between the online and offline classes.

**5.8 T-test:** In statistics, various tests are used to compare different samples or groups and draw conclusions about populations.

- Statistical tests analyze the likelihood of obtaining observed data under specific assumptions.
- Provide a framework for assessing evidence in support or against a hypothesis.
- Starts by formulating a null hypothesis ( $H_0$ ) and an alternative hypothesis ( $H_a$ ).
- Named after William Sealy Gosset's Student's t-distribution.
- Used in inferential statistic test to determine significant difference between means of two groups.
- Used when data is normally distributed and population variance is unknown.
- Assesses whether observed difference is statistically significant or due to random variation.

### 5.8.1 Assumptions in T-test

- **Independence:** Observations within each group must be independent, preventing one observation from influencing another.
- **Normality:** Data within each group should be approximately normally distributed, crucial for small sample sizes.
- **Homogeneity of Variances:** The variances of the two groups should be equal, ensuring similar value spread.
- **Absence of Outliers:** No extreme outliers in the data, as they can disproportionately influence results, especially when sample sizes are small.

### 5.8.2 Prerequisites for T-Test

- T-Test is a statistical method used to compare means of two groups.
- It is a parametric test, making certain assumptions about the data.
- **Hypothesis Testing:** A statistical method used to make inferences about a population based on a sample of data.
- **P-value:**
  - The probability of observing a test statistic given that the null hypothesis is true.
  - Small p-value suggests the observed data is unlikely to have occurred by random chance alone, leading to the rejection of the null hypothesis.
- **Degree of Freedom (df):**
  - Represents the number of values in a calculation that are free to vary.

- Calculated as the total sample size minus 1 in a t-test.

$$df = \sum n_s - 1$$

- Reflects the number of values in the sample that are free to vary after estimating the sample mean.

$$df = (nA - 1) + (nB - 1)$$

- **Significance Level:**
  - Predetermined threshold used to decide whether to reject the null hypothesis.
  - Commonly used levels are 0.05, 0.01, or 0.10.
- **T-Statistic:**
  - Measures the difference between the means of two groups relative to the variability within each group.
  - Calculated as the difference between the sample means divided by the standard error of the difference.
- **T-Distribution:**
  - also known as the Student's t-distribution, is a probability distribution with thicker tails than the normal distribution.

### 5.8.3 Statistical Significance

- Statistical significance determined by comparing p-value to chosen significance level.
- P-value less than or equal to significance level indicates statistical significance, rejecting null hypothesis.
- P-value greater than significance level indicates insufficient evidence to reject null hypothesis.
- T-test assesses significant difference between groups, considering group variability.
- P-value from t-test compared to significance level to decide null hypothesis.

### 5.8.4 Types of T-tests

There are three types of t-tests, and they are categorized as dependent and independent t-tests.

- **One sample t-test test:** The mean of a single group against a known mean.
- **Two-sample t-test: It is further divided into two types:**
- **Independent samples t-test:** compares the means for two groups.
- **Paired sample t-test:** compares means from the same group at different times (say, one year apart).

## 5.9 Chi-square test

- Crucial for analyzing associations in categorical data.
- Aids in understanding factor relationships.
- Explains types, implementation steps, and feature selection role.
- Exemplified using Python code on Iris dataset.

The Chi-Square test is a non-parametric statistical test that determines the significant association between two categorical variables. It compares observed and expected frequencies within a contingency table, aiding in feature selection and ensuring the association between the sample and the population. It belongs to the continuous probability distribution family.

$$\chi_c^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

The Chi-Square Distribution is a statistical probability distribution, derived from the sum of squares of independent standard normal random variables, used in hypothesis testing and confidence interval calculations, especially with non-normally distributed data.

### Key terms used in Chi-Square test

#### Degrees of freedom

**Observed values:** Actual data collected

**Expected values:** Predicted data based on a theoretical model in chi-square test.

$$E_{ij} = \frac{R_i \times C_j}{N}$$

**Contingency table:** A contingency table, also known as a cross-tabulation or two-way table, is a statistical table that displays the distribution of two categorical variables.

### 5.9.1 Chi-Square Test Types and Applications

**Chi-Square Test for Independence:** Assesses significant association between two categorical variables.

- Used when counts of values for two nominal or categorical variables are available.
- Requires independence of observations and a large sample size.
- Example: Investigating relationship between online shopping preferences and payment methods.

**Chi-Square Goodness-of-Fit Test:** Used in statistical hypothesis testing to determine if a variable is likely from a given distribution.

- Used when value counts for categorical variables are available.

- Can determine if data values are a representative sample of the entire population or if they fit the hypothesis well.
- Example: Testing fairness of a six-sided die.

### 5.9.2 Chi-Square Test: A Comprehensive Analysis

- Widely used across diverse fields for categorical data analysis.
- Tests the independence of two categorical variables.
- Helps understand relationships between factors like gender, preferences, product categories, and purchasing behaviors.
- Appreciated for its simplicity and ease of application.
- Provides insights into patterns and associations within categorical data.
- Applicable in fields like genetics, market research, quality control, and social sciences.
- Assesses data conformity to expected values, enhancing its role in statistical analysis.

### 5.9.3 Chi-square Test Procedure

- **Define Null Hypothesis (H0):** No significant association between two categorical variables.
- **Create Contingency Table:** Displays frequency distribution of two variables.

Find Expected Values: 
$$E_{ij} = \frac{R_i \times C_j}{N}$$

- **Calculate Chi-Square Statistic:**

Calculate Degrees of Freedom: 
$$df = (m-1)(n-1)$$

- **Accept or Reject Null Hypothesis:** Compare chi-square statistic to critical value for chosen significance level (e.g., 0.05).

### 5.9.4 Chi-square Test for Feature Selection

- Used for categorical features in a dataset.
- Calculates Chi-square between each feature and target.
- Selects the number of features with best Chi-square scores.
- Features with significant dependencies with target variable are important for prediction.
- Feature selection, also known as attribute selection, extracts relevant features for machine learning algorithms.
- Large number of irrelevant features increases training time and risk of overfitting.

**Example:** examining "income level" and "subscription status" to determine relevance for predicting subscription status.

## 5.10 F-Test

The F test is a statistical method used in hypothesis testing to determine the equality of variances between two populations or samples. It can be one-tailed or two-tailed and can be calculated using the f statistic.

### F-Distribution in Statistical Hypothesis Testing

- Used in comparing variances or testing means across populations.
- Defined by two degrees of freedom parameters: df1 (numerator) and df2 (denominator).
- Assesses equality of variances or means in statistical analyses like ANOVA or regression.
- Determines rejection zone in hypothesis testing based on critical values from F-distribution.
- Found when two independent chi-square variables are split by their degrees of freedom.

**Formula for F-distribution:**

$$f\text{-value} = \frac{\text{sample1}/df1}{\text{sample2}/df2}$$

- The independent random variables, Samples 1 and 2, have a chi-square distribution.
- The related samples' degrees of freedom are denoted by df1 and df2.

### Degree of Freedom in Chi-square Variables

- Represents number of observations used to calculate chi-square variables.
- Determines F-distribution's shape.
- Right-skewed distribution with longer tail on right side.
- As degrees increase, F-distribution becomes symmetric and bell shape.

### F-Test Overview

- Statistical technique determining equal variances of two samples or populations.
- Requires both samples and populations to be independent and fit into an F-distribution.
- Null hypothesis can be rejected if F test results are statistically significant.

### F-Test Formula using two covariances



$$F_{calc} = \frac{\sigma_1^2}{\sigma_2^2}$$

Here,

- $F_{calc}$  = Critical F-value.
- $\sigma_1^2$  &  $\sigma_2^2$  = variance of the two samples.

$$df = n_S - 1$$

Here,

- $df$  = Degrees of freedom of the sample.
- $n_S$  = Sample size.

### Steps to calculate F-Test

- **Step 1:** Use Standard deviation ( $\sigma$ ) and find variance ( $\sigma^2$ ) of the data. (if not already given)
- **Step 2:** Determine the null and alternate hypothesis.
  - $H_0$ : no difference in variances.
  - $H_1$ : difference in variances.
- **Step 3:** Find  $F_{calc}$  using Equation 1 (F-value).  
NOTE: While calculating  $F_{calc}$ , divide the larger variance with small variance as it makes calculations easier.
- **Step 4:** Find the degrees of freedom of the two samples.
- **Step 5:** Find  $F_{table}$  value using  $d_1$  and  $d_2$  obtained in Step-4 from the F-distribution table. Take learning rate,  $\alpha = 0.05$  (if not given).